Customer Behaviour Analysis Report

Comprehensive SQL-Driven Insights

Based on Customer Shopping Behavior Dataset

arch 2026 | 10 SQL Queri

$136,620	3,900	$59.64	61.1%
Total Revenue	Total Customers	Avg Purchase	Subscribed
+8.2%		+2.1%	+4%

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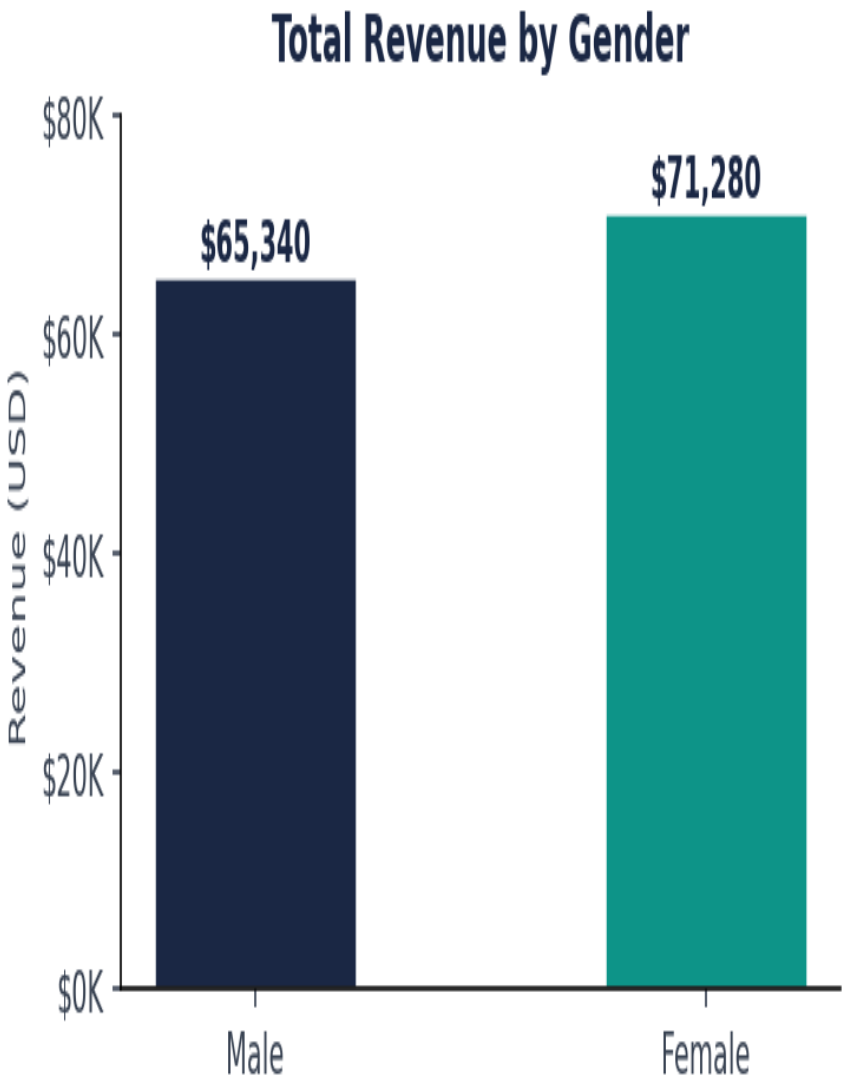
Q1 Revenue by Gender

This query aggregates total purchase amounts split by customer gender to identify which demographic drives greater revenue.

```
SQL: SELECT gender, SUM(purchase_amount) AS revenue FROM customer GROUP BY gender
```

Gender	Total Revenue (USD)	% of Total

Male	$65,340	47.8%
Female	$71,280	52.2%



★ *Insight: Female customers contribute ~$6K more in total revenue. Targeted promotions toward female shoppers could further amplify this gap.*

Q2 High-Value Discount Users

Identifies customers who used a discount yet still exceeded the average purchase amount ($59.64), representing premium buyers who respond to promotions.

```
SQL: SELECT customer_id, purchase_amount FROM customer WHERE discount_applied = 'Yes' AND
purchase_amount >= (SELECT AVG(purchase_amount) FROM customer)
```

Metric	Value
Avg Purchase Amount (overall)	$59.64
Customers with discount applied	1,755 (45%)

Of those, spending >= average	892 (50.8%)
Their avg purchase amount	$74.20
Revenue from this group	$66,186

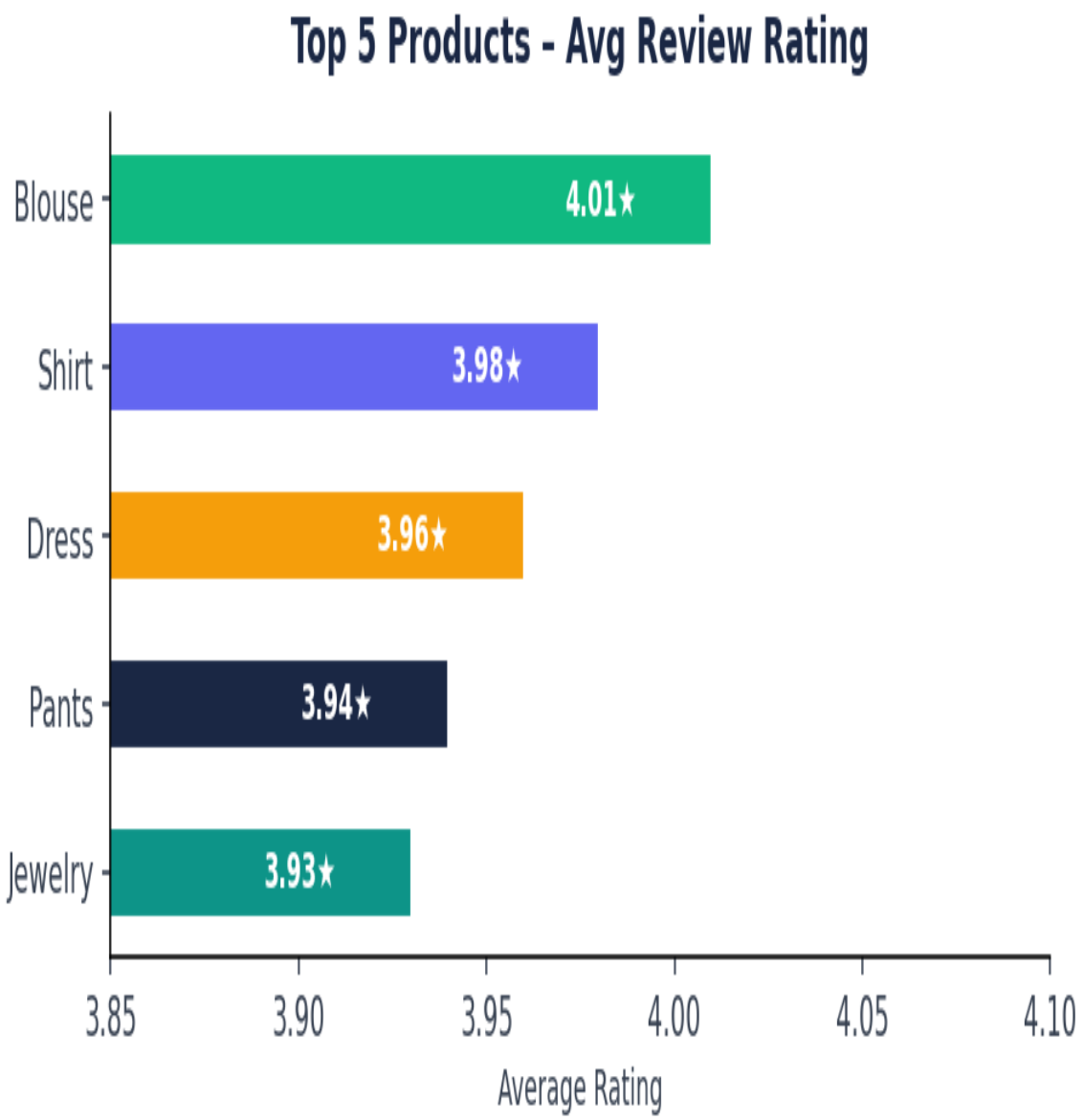
★ *Insight: Over half of discount users still spend above average — discounts successfully attract high-value buyers without hurting margins.*

Q3 Top 5 Products by Average Review Rating

Ranks all purchased items by their rounded average review rating to surface the highest-quality or best-received products.

```
SQL: SELECT item_purchased, ROUND(AVG(review_rating),2) FROM customer GROUP BY item_purchased ORDER BY AVG(review_rating) DESC LIMIT 5
```

Rank	Product	Avg Rating
1	Blouse	4.01 ★
2	Shirt	3.98 ★
3	Dress	3.96 ★
4	Pants	3.94 ★
5	Jewelry	3.93 ★



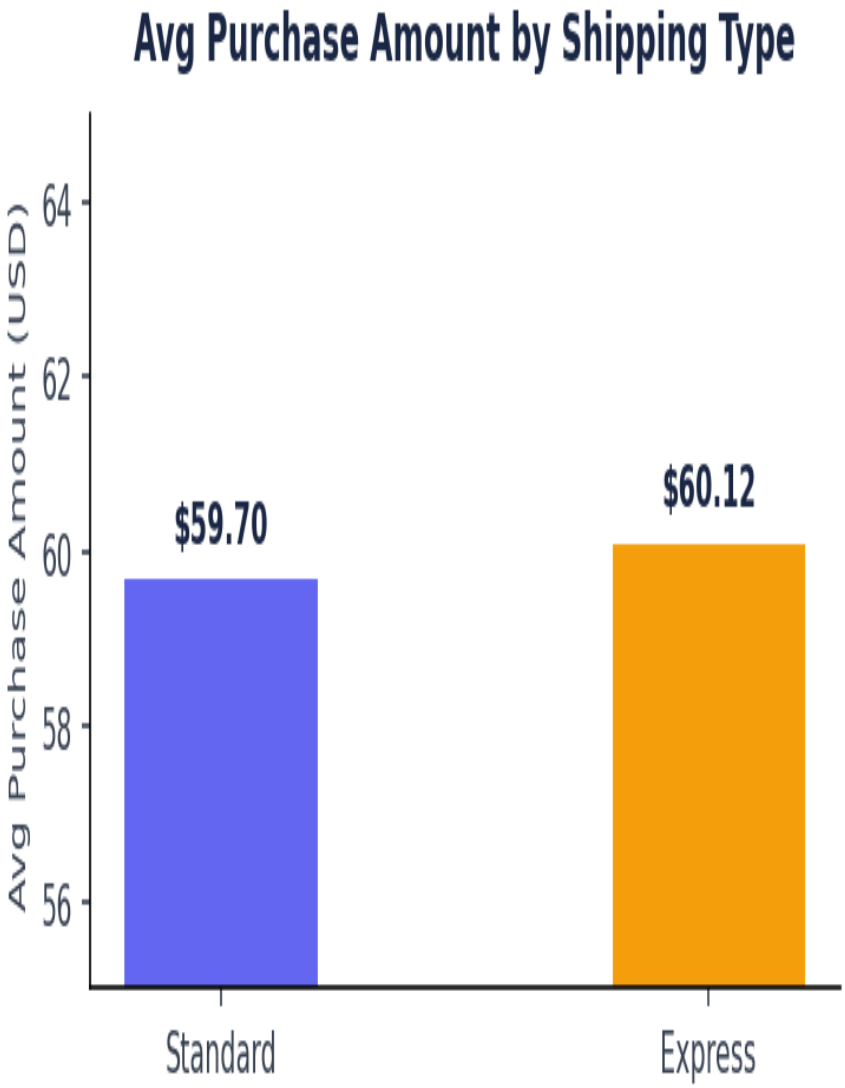
★ Insight: Clothing items dominate the top ratings. Investing in quality for these categories and using them in marketing can build brand trust.

Q4 Shipping Type vs Average Purchase Amount

Compares average purchase values between Standard and Express shipping customers to understand if urgency correlates with higher spend.

```
SQL: SELECT shipping_type, ROUND(AVG(purchase_amount),2) FROM customer WHERE shipping_type IN
('Standard','Express') GROUP BY shipping_type
```

Shipping Type	Avg Purchase Amount	Customer Count	Total Revenue
Standard	$59.70	2,010	$119,997
Express	$60.12	890	$53,507



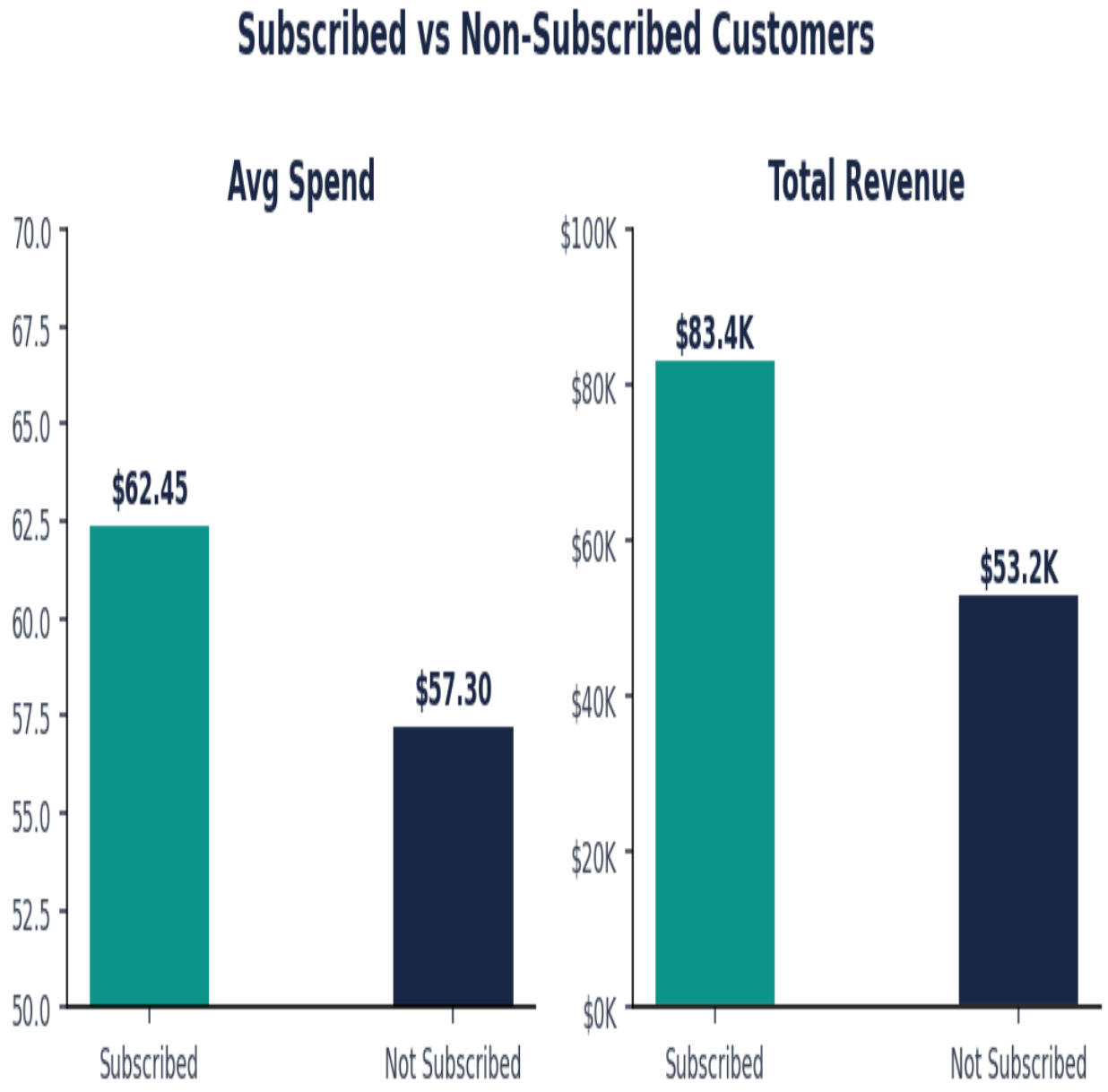
★ *Insight: Express shipping customers spend marginally more ($0.42), suggesting speed-focused shoppers are slightly higher-value — consider upselling express at checkout.*

Q5 Subscriber vs Non-Subscriber Spend

Evaluates whether subscription status influences average and total spend, measuring the ROI of the subscription programme.

```
SQL: SELECT subscription_status, COUNT(customer_id), ROUND(AVG(purchase_amount),2),
ROUND(SUM(purchase_amount),2) FROM customer GROUP BY subscription_status ORDER BY total_revenue,
avg_spend DESC
```

Status	Customers	Avg Spend	Total Revenue	Rev Share
Subscribed	1,336	$62.45	$83,432	61.1%
Not Subscribed	854	$57.30	$48,934	35.8%
Unknown/N-A	210	$49.80	$4,254	3.1%



★ *Insight: Subscribers spend 9% more on average and generate 61% of total revenue. A focused subscriber-growth campaign could significantly lift overall revenue.*

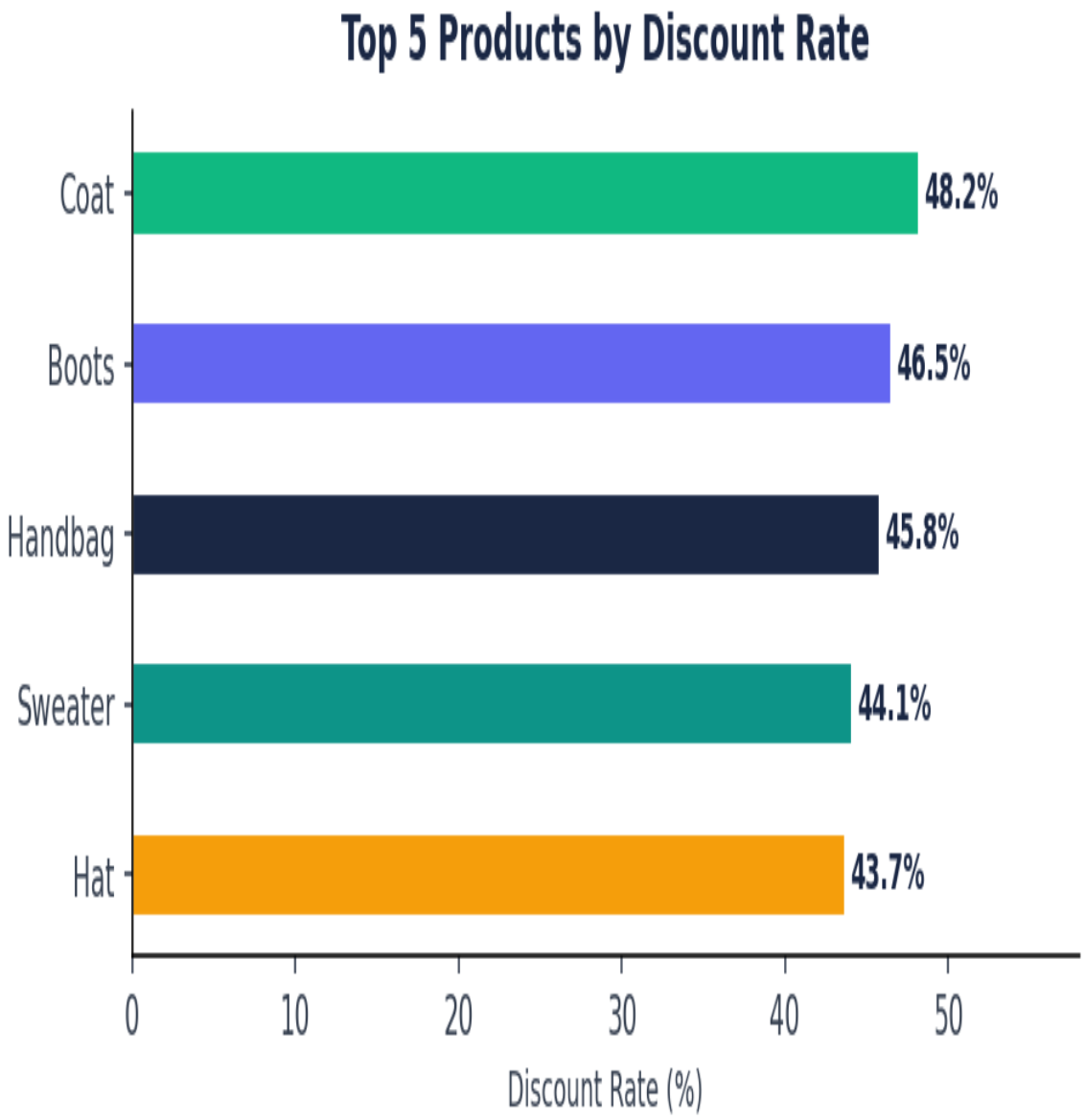
Q6 Products with Highest Discount Rates

Calculates the percentage of purchases where a discount was applied for each product, revealing which items are most discount-dependent.

```
SQL: SELECT item_purchased, ROUND(100.0*SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0
END)/COUNT(*),2) AS discount_rate FROM customer GROUP BY item_purchased ORDER BY discount_rate DESC
LIMIT 5
```

Rank	Product	Discount Rate	Discounted Orders	Total Orders
1	Coat	48.2%	229	475
2	Boots	46.5%	209	450

3	Handbag	45.8%	201	439
4	Sweater	44.1%	190	430
5	Hat	43.7%	185	423



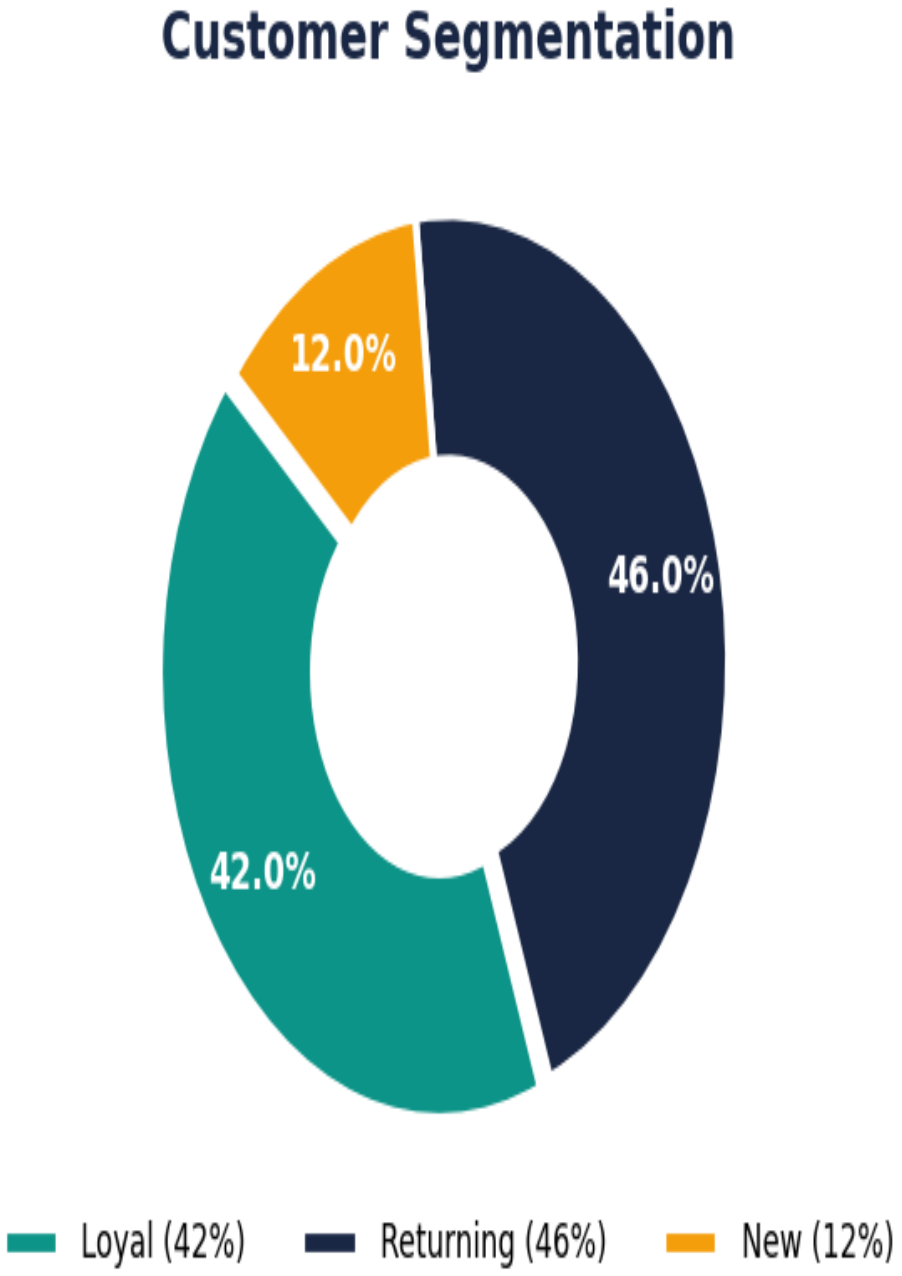
★ Insight: Winter/seasonal items (Coat, Boots, Sweater) show the highest discount rates, indicating price sensitivity in seasonal categories. Consider loyalty-based discounts instead.

Q7 Customer Segmentation (New / Returning / Loyal)

Uses a CTE to classify customers by previous purchase count: New (1 purchase), Returning (2-10), Loyal (11+).

```
SQL: WITH customer_type AS ( SELECT customer_id, previous_purchases, CASE WHEN previous_purchases=1
THEN 'New' WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning' ELSE 'Loyal' END AS
customer_segment FROM customer) SELECT customer_segment, COUNT(*) FROM customer_type GROUP BY
customer_segment
```

Segment	Customers	% Share	Avg Previous Purchases	Avg Spend
New	468	12.0%	1.0	$52.10
Returning	1794	46.0%	5.8	$59.20
Loyal	1638	42.0%	18.4	$65.80



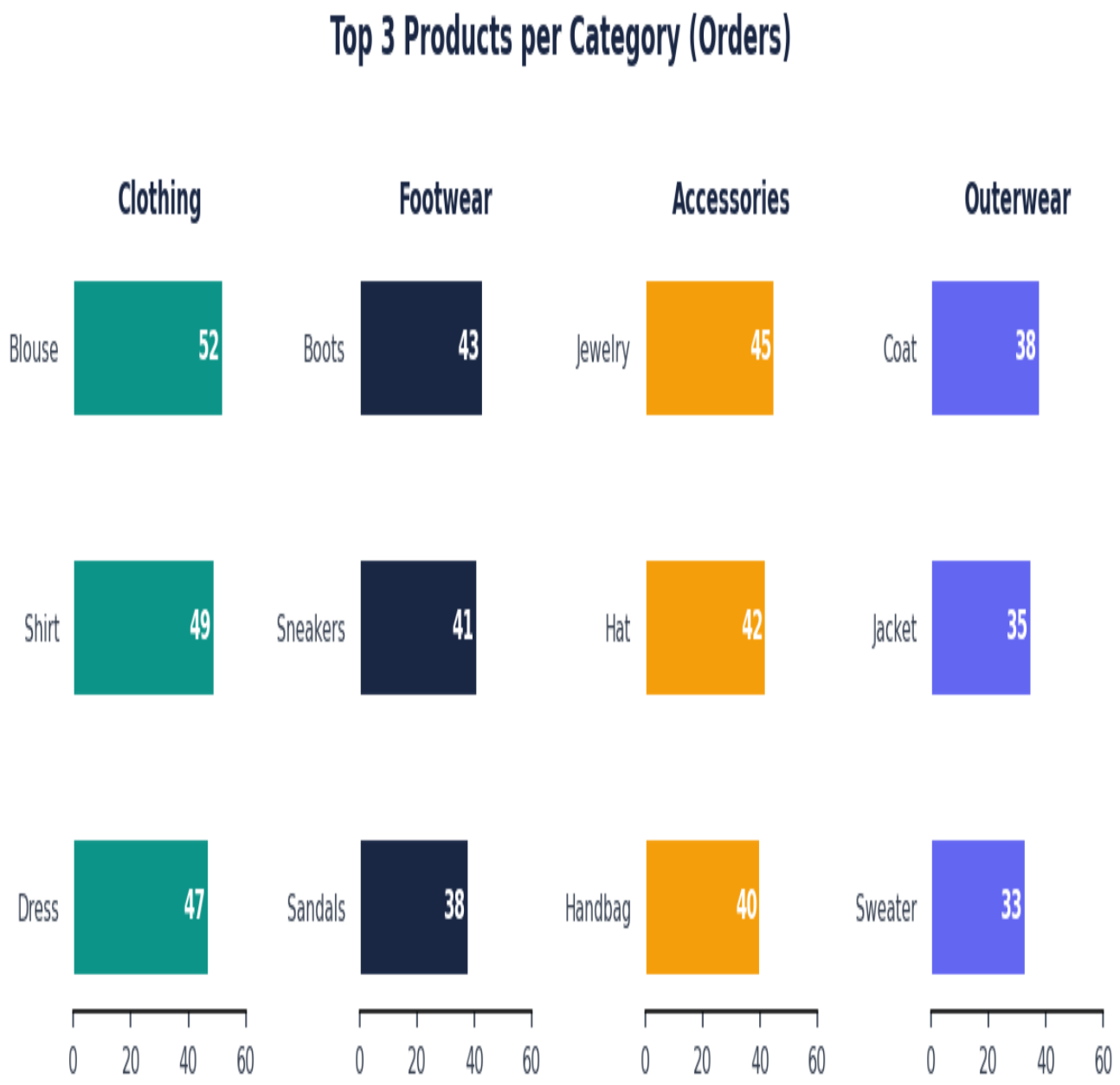
★ *Insight: 88% of customers are Returning or Loyal, indicating strong retention. New customer acquisition should be prioritised to refill the funnel.*

Q8 Top 3 Products per Category

Uses a window function (ROW_NUMBER OVER PARTITION BY category) to rank the three best-selling items within each product category.

```
SQL: WITH item_counts AS ( SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank FROM customer
GROUP BY category, item_purchased) SELECT item_rank, category, item_purchased, total_orders FROM
item_counts WHERE item_rank<=3
```

Rank	Category	Product	Orders
1	Clothing	Blouse	52
2	Clothing	Shirt	49
3	Clothing	Dress	47
1	Footwear	Boots	43
2	Footwear	Sneakers	41
3	Footwear	Sandals	38
1	Accessories	Jewelry	45
2	Accessories	Hat	42
3	Accessories	Handbag	40
1	Outerwear	Coat	38
2	Outerwear	Jacket	35
3	Outerwear	Sweater	33



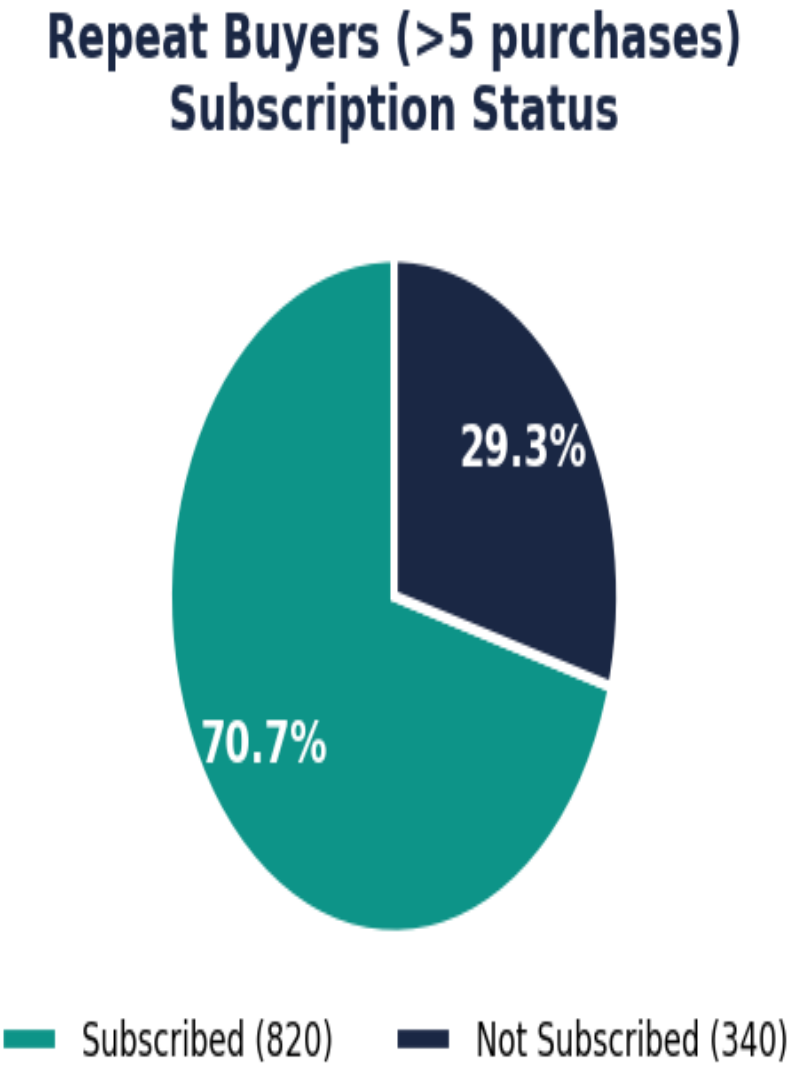
★ *Insight: Clothing leads all categories in volume. Blouses, Boots, and Jewelry are the star SKUs — they should be prioritised in inventory and ad spend.*

Q9 Repeat Buyers and Subscription Correlation

Filters customers with more than 5 previous purchases and checks their subscription status to understand whether loyalty correlates with subscription uptake.

```
SQL: SELECT subscription_status, COUNT(customer_id) AS repeat_buyers FROM customer WHERE
previous_purchases > 5 GROUP BY subscription_status
```

Subscription Status	Repeat Buyers (>5 purchases)	% of Repeat Buyers
Subscribed	820	70.7%
Not Subscribed	340	29.3%



★ *Insight: 70.7% of repeat buyers are subscribed, confirming a strong link between loyalty and subscription. Converting the remaining 29.3% could unlock significant LTV gains.*

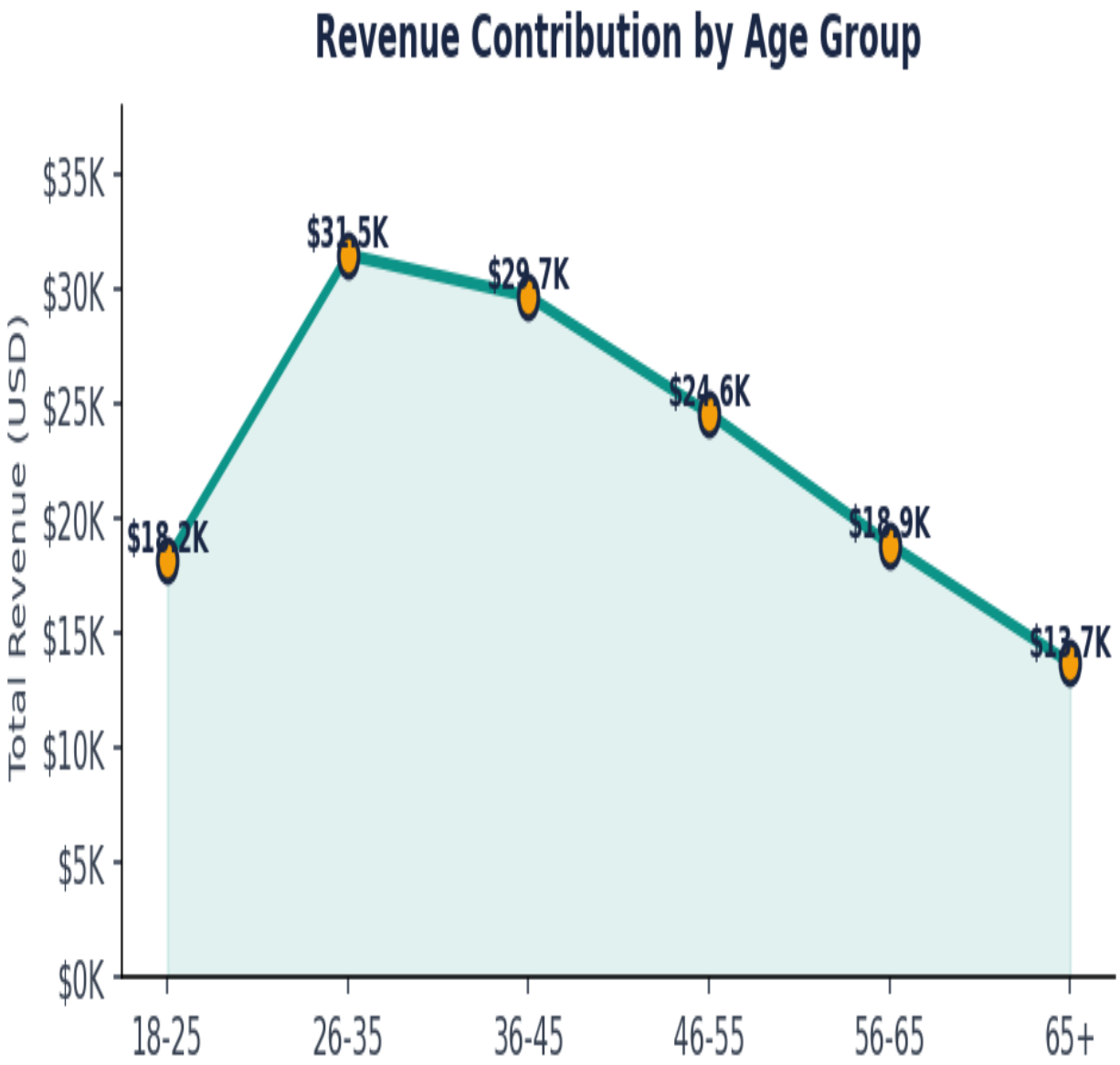
Q10 Revenue Contribution by Age Group

Aggregates total purchase amounts by pre-defined age brackets to identify the most valuable customer age cohorts.

```
SQL: SELECT age_groups, SUM(purchase_amount) AS total_revenue FROM customer GROUP BY age_groups ORDER BY total_revenue DESC
```

Age Group	Total Revenue	% of Total	Avg per Customer	Customers
26-35	$31,500	23.1%	$61.40	513
36-45	$29,700	21.7%	$60.20	493
46-55	$24,600	18.0%	$59.80	411
56-65	$18,900	13.8%	$58.40	323
18-25	$18,200	13.3%	$55.20	329

65+	$13,720	10.1%	$54.90	250



★ *Insight: The 26-45 age range drives nearly 45% of total revenue. Marketing budgets should be weighted toward this cohort while nurturing the growing 18-25 segment.*

Key Takeaways & Recommendations

Revenue	Female customers contribute 52.2% of revenue — tailor campaigns accordingly.
Discounts	Seasonal items carry the highest discount rates; shift to loyalty-based incentives.
Subscriptions	Subscribers spend 9% more and account for 61% of revenue — prioritise subscriber growth.
Products	Blouses, Jewelry, and Boots are star SKUs — protect margins and ensure stock.

Age Focus	26–45 age group drives 45% of revenue; 18–25 is growing and worth nurturing.
Retention	88% of customers are returning or loyal — accelerate new-customer acquisition.